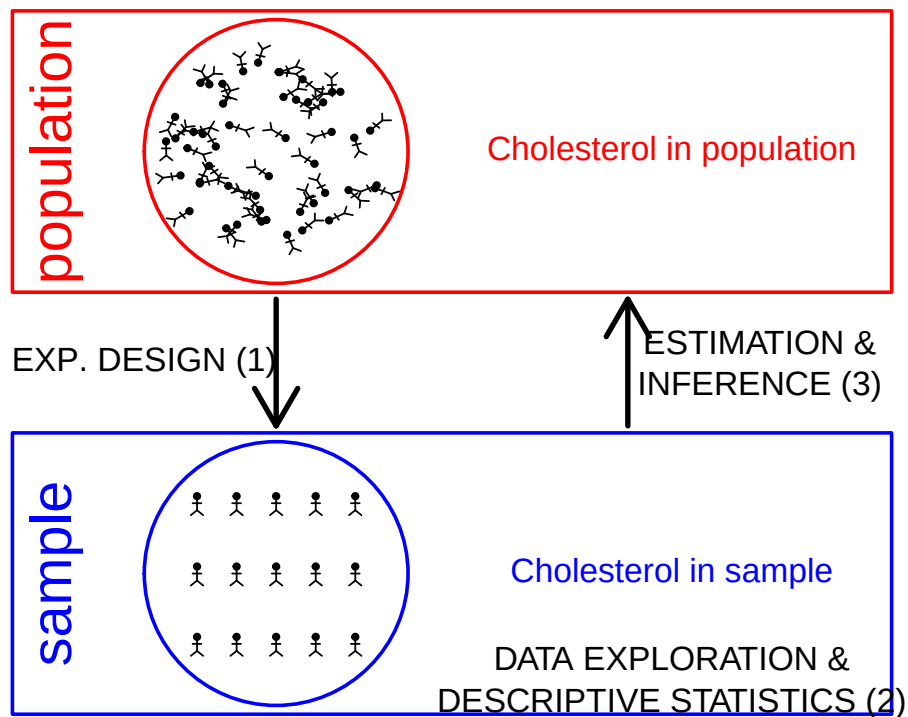


3. Some concepts on experimental design

statOmics, Ghent University (<https://statomics.github.io>)

Lieven Clement



Need for a good control

- A good control group is crucial.
- To assess the effect of an intervention, we need to compare a test and control group.

- This is often not possible in a pretest/post-test design: e.g. effect before and after administering a drug without the use of a placebo group.
 - Groups in an observational study are often not comparable: advanced statistical methods are required to draw causal conclusions.
 - Double blinding
 - We have to be aware of confounding!
 - Randomized studies: random assignment of subjects in the study to the different treatment arms → comparable groups.
-

Randomization

- Randomization completely at random (no systematic allocation).

Simple Randomization

- Can lead to differences in the number of experimental units in each treatment arm
 - in 5% of the cases we might observe an imbalance of
 - of at least 60:40 in a study with 100 subjects, and
 - of at least 531:469 in a study with 1000 subjects.
 - This imbalance is not problematic, but causes a loss in precision.
-

Balanced Randomization

- Equal numbers of each treatment are assigned to a block of 2 or 4 patients.
 - (1) AB, (2) BA
 - (1) AABB, (2) ABAB, (3) ABBA, (4) BABA, (5) BAAB, (6) BBAA
- Balanced Randomization ensures $\pm$ the same number of people in the control and the treatment arm of the experiment.
- Does not make that we have an equal number of males with and without the treatment, etc.

- In small studies, it is possible that the groups are unbalanced in other characteristics (e.g. gender, race, age ...)
- This is not problematic because it occurs at random, but, again it causes a loss in precision.

Stratified randomization

- The imbalance according to for instance gender can be avoided using stratified Randomization: balanced randomization per stratum

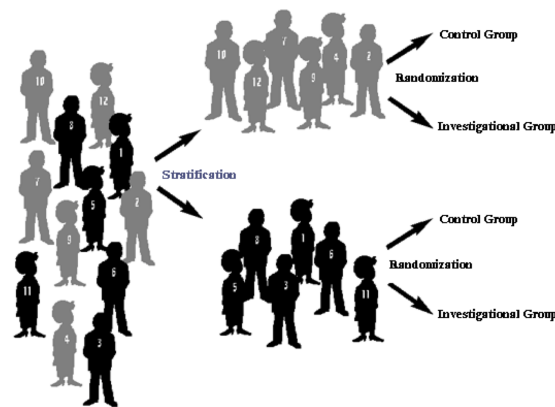


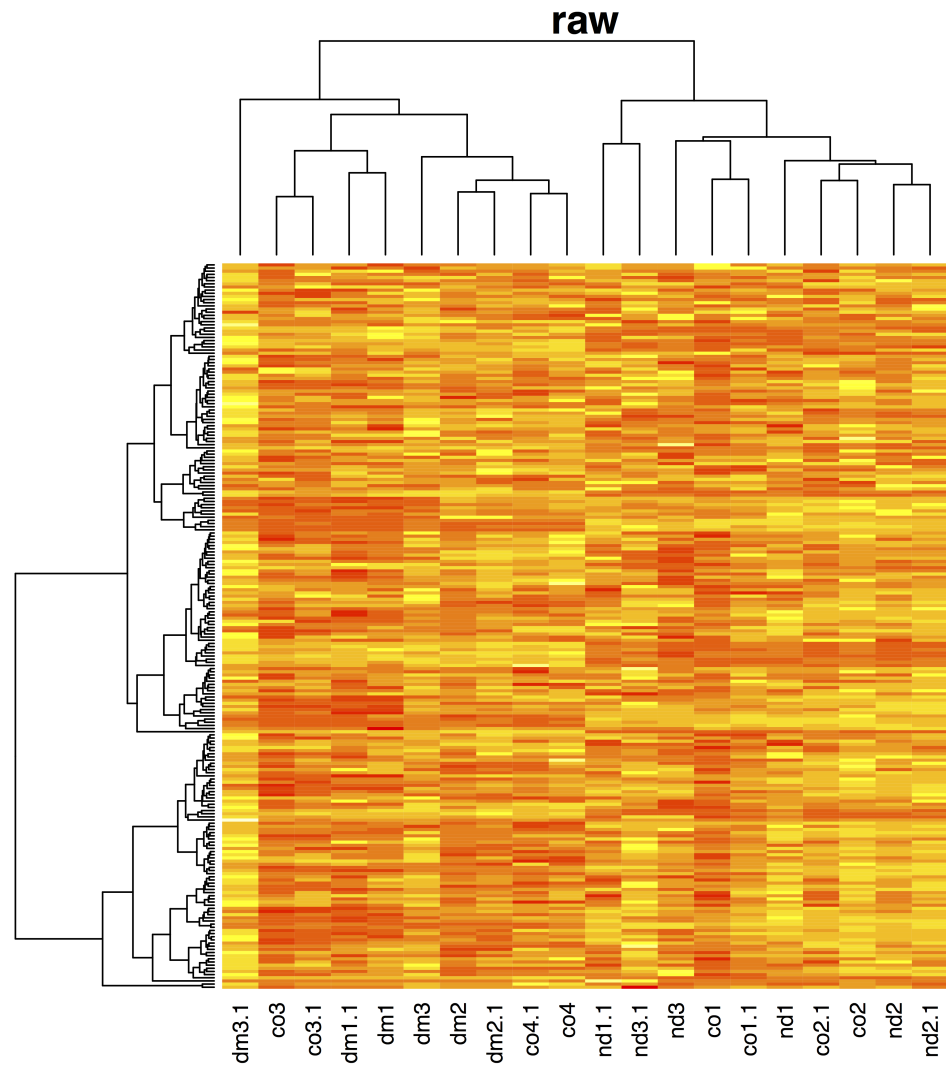
Figure 1: Stratified Randomization

Sample size

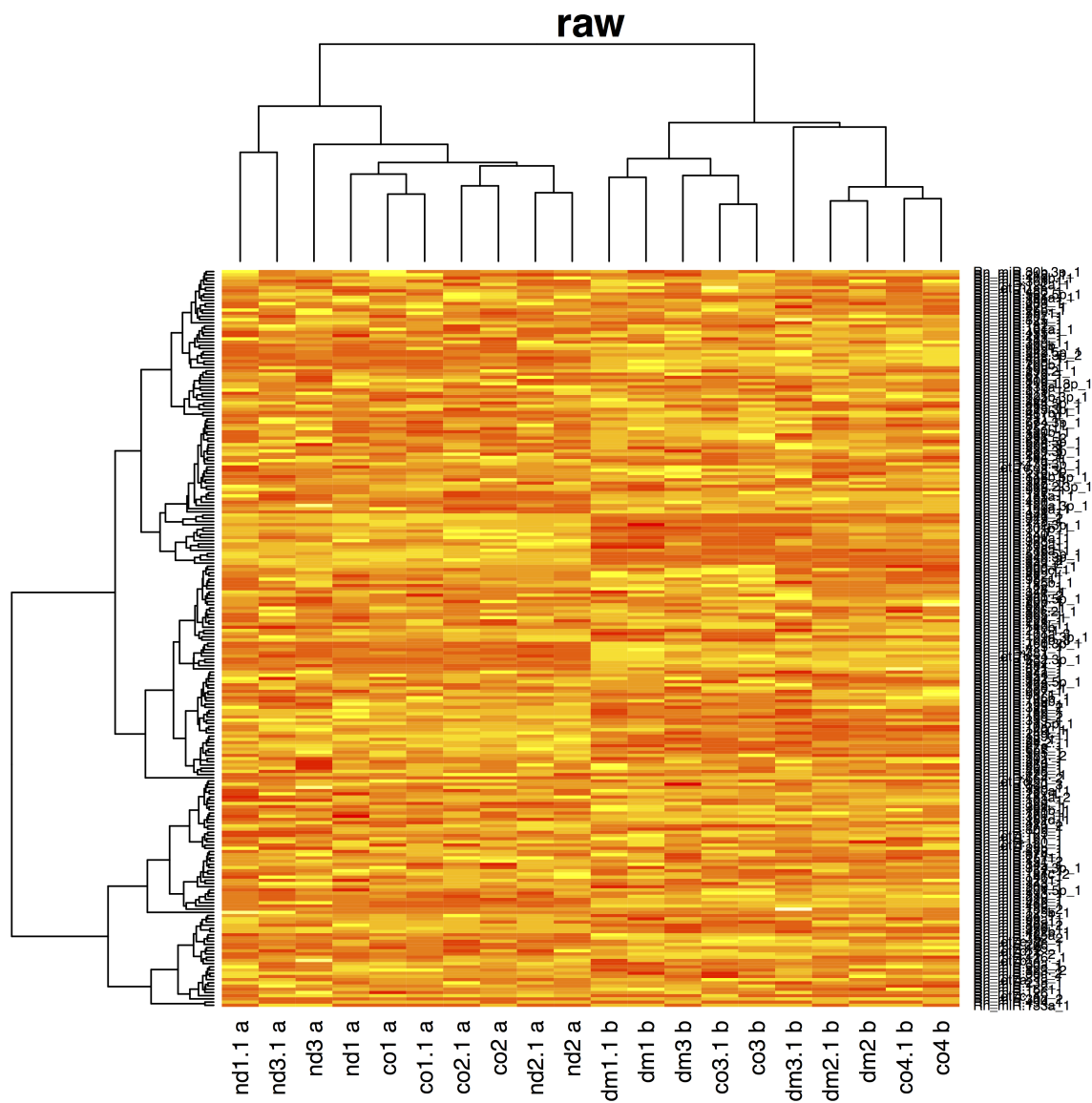
- The sample size and the design are crucial.
- The larger the sample size, the more precise the results.

Bad design example

- dm: diabetic medium, nd: non diabetic medium, co: control



- 4 bio-reps, 2 techreps/biorep
- dm: diabetic medium, nd: non diabetic medium, co: control
- 4 bio-reps, 2 techreps/biorep, 2 plates A & B
- Treatment and plate almost entirely confounded



Wrap-up

- Sample size is very important.
- To assess the effect of a treatment, we should compare comparable and representative groups of subjects with and without the treatment (a good control!).

- In observational studies, the researcher cannot choose the treatment. It was the patient or their MD who had chosen it
- In experimental studies, the researcher assigns the treatment.
- Confounding can be avoided via randomization.
- We can also correct for confounding in the statistical analysis for the confounders that have been registered.